

Zoe Hackshaw

Curriculum Vitae

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EDUCATION	Ph.D. in Astronomy , The University of Texas at Austin Advisor: Dr. Keith Hawkins Expected June 2027	
	M.A. in Astronomy , The University of Texas at Austin Thesis: <i>Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE</i> Advisor: Dr. Keith Hawkins	Dec. 2024
	B.S. in Astrophysics , The University of Florida <i>summa cum laude</i> Thesis: <i>J0010-1735—A Stellar Journey from an R-process Enhanced Ultra Faint Dwarf Galaxy to the Milky Way Halo</i> Advisor: Dr. Rana Ezzeddine	May 2021
APPOINTMENTS	Graduate Student Researcher Department of Astronomy, The University of Texas at Austin Advisor: Dr. Keith Hawkins	2022 - Present
	Pre-Doctoral Fellow Center for Computational Astrophysics, Flatiron Institute Advisors: Dr. Adrian Price-Whelan, Dr. Carrie Filion, Dr. Aida Behmard	Jan. 2026 - May 2026
	Undergraduate Student Researcher Department of Astronomy, The University of Florida Dr. Rana Ezzeddine	2019 - 2021
SELECT HONORS AND AWARDS	Center for Computational Astrophysics Pre-Doctoral Fellowship Flatiron Institute, Simons Foundation	2026
	David L. Lambert Graduate Fellowship Department of Astronomy, The University of Texas at Austin	2024
	Senior Thesis Highest Honors Department of Astronomy, The University of Florida	2021
PUBLICATIONS	First Author	
	🖱️ Clickable link 2. <i>A Detailed Chemical Analysis of the Red Giant Orbiting Gaia BH3: From Lithium to Thorium</i> Hackshaw Z. , Hawkins K., and Manea C. 2025 ApJ , DOI 10.3847/1538-4357/ae1617 1. <i>[X/Fe] Marks the Spot: Mapping Chemical Azimuthal Variations in the Galactic Disk with APOGEE</i> Hackshaw Z. , Hawkins K., Filion C., Horta D., Laporte C. F. P., Carr C., Price-Whelan A. M et al 2024 ApJ , DOI 10.3847/1538-4357/ad900e	

Contributing Author

6. *An Analytic Model for Stellar Metallicity Gradient Residuals in Cold, Phase-Mixed Galactic Disks*
Modak S., **Hackshaw Z.**, and Hawkins K. 2026, **ApJL**, DOI 10.48550/arXiv.2606.14850
5. *Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk: Insights from Kinematic Ridges with Gaia*
Jurado C., Hawkins K., Hunt J. A. s., **Hackshaw Z.**, et al. 2025, **ApJ**, DOI 10.48550/ arXiv.2601.02494
4. *Exploring the History of Stellar Mergers with Chemistry: Examining the Origins of Massive α -enriched Stars Using Carbon Isotope Ratios*
Maas Z. G., Hawkins K., Gerber J. M., **Hackshaw Z.**, Manea C., et al. 2025, **ApJ**, DOI 10.48550/arXiv.2509.08900
3. *Optical Spectroscopy Reveals Hidden Neutron-capture Elemental Abundance Differences among APOGEE-identified Chemical Doppelgängers*
Manea C., Ness M., Hawkins K., Zeimann G., Hogg D. W., Filion C., Griffith E., Johnston K., Casey A., **Hackshaw Z.**, et al. 2025, **ApJ**, DOI 10.48550/arXiv.2508.16717
2. *Discovery of a Metal-poor Red Giant Star with the Highest Ultralithium Enhancement*
Kowkabany J., Ezzeddine R., Charbonnel C., Roederer I. U., Wang E. X., Li Y., **Hackshaw Z.**, et al 2024, **ApJ**, DOI 10.3847/1538-4357/ad6004/ arXiv.2209.02184
1. *Uranium Abundances and Ages of r-process Enhanced Stars with Novel U II Lines*
Shah S., Ezzeddine R., Ji A., Hansen T., Catelan M., **Hackshaw Z.**, et al 2023, **ApJ**, DOI 10.3847/1538-4357/acb8af

SELECT SCIENTIFIC PRESENTATIONS

👉 Clickable link

Cool Stars, Stellar Systems and the Sun 23, Poster University of Tokyo, ABC, ISAS-JAXA	June 2026
Pre-Doctoral Symposium, Talk Center for Computational Astrophysics; Flatiron Institute	May 2026
Astronomy Seminar, Talk Columbia Univeristy	March 2026
s,i,r- Element Nucleosynthesis (sirEN) Conference, Poster Italian National Institute for Astrophysics	June 2025
Milky Way Assembly Tale Conference, Poster Istituto Nazionale Di Astrofisica	June 2024
SDSS-V Collaboration Meeting, Contributing Talk Sloan Digital Sky Survey-V	Aug. 2023
Stellar Properties from NLTE Radiative Transfer, Contributing Talk IReNA, Kavli Institute for Cosmological Physics, and the University of Chicago	May 2023
AAS Meeting, Poster American Astronomical Society	Jan. 2021

SELECT OBSERVING EXPERIENCE

PI, Pollution of FGK White Dwarf Companions, 7 nights Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory	Oct. 2024
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	PI, <i>Chemical Analysis of Gaia BH3</i> , 7 nights <i>Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory</i>	June 2024
	PI, <i>Cosmochronometry with Asteroseismic Ages</i> , 7 nights <i>Tull Coudé Spectrograph, 2.7m HJST, McDonald Observatory</i>	Feb. 2024
COMPUTER SKILLS	<i>Languages: Python, Julia, C++</i> <i>Software: Turbospectrum, BACCHUS, Spectroscopy Made Hard, Git, GitHub, LaTeX</i>	
TEACHING & MENTORSHIP	REU McDonald Observatory Trip Lead <i>McDonald Observatory, The University of Texas at Austin</i>	July 2026
	Graduate-Undergraduate Mentorship in astronoMY Mentor <i>The University of Texas at Austin</i>	2024 - Present
	Research Mentor, Students: Abigail Skelton and Nachiket Yadav <i>The University of Texas at Austin</i>	2024
	Teaching Assistant, AST 325K: Stellar Astronomy <i>The University of Texas at Austin</i>	2023
OUTREACH & SCIENCE COMM. 👉 Clickable link	AAS Journal Author Series, Invited Talk <i>American Astronomical Society</i>	June 2026
	Astronomy Club Guest Lecture, Invited Talk <i>Davidson Academy Online</i>	March 2026
	Fourth Grade Astronomy Lesson, Invited Talk <i>Eagle Ridge Elementary</i>	Feb. 2026
	Live Deep Sky Tour Stream, Invited Talk <i>McDonald Observatory</i>	Dec. 2025
	Future in STEM Career Seminar, Invited Talk <i>Lynn University</i>	Oct. 2025
	Astronomy Day Distance Learning Presentation, Invited Talk <i>McDonald Observatory</i>	Sept. 2025
	Explore our Solar System Teacher Workshop, Invited Co-Lead <i>McDonald Observatory</i>	July 2025
	Astronomy on Tap ATX Presentation, Contributing Talk <i>Astronomy on Tap Austin, TX</i>	Feb. 2025
	UT STEM Girl Day, Contributing Activity Lead <i>The University of Texas at Austin</i>	Feb. 2023 - 2025
	Astronomy at the Library Presentation, Contributing Talk <i>Williamson County Astronomy Club</i>	Nov. 2024
	Board of Visitors Summer Meeting, Invited Talk <i>McDonald Observatory and the University of Texas at Austin</i>	July 2024
	Girl Scout Space Badge Presentation, Invited Talk <i>Daisy Troop 41119, Austin, TX</i>	April 2022
LEADERSHIP & EXTERNAL ACTIVITIES	Social Media and Marketing Manager <i>Astronomy on Tap, Austin, TX Chapter</i>	2023 - Present
	Prospective Graduate Student Visit Lead <i>Department of Astronomy, The University of Texas</i>	2023-2025
	Assistant Director of Mini Marathons <i>Morale Team for Dance Marathon at the University of Florida</i>	2020 - 2021

Vice President of Chapter Relations and Standards
Alpha Chi Omega, Gamma Iota Chapter

2018 - 2019

**PROFESSIONAL
EXPERIENCE**

Engineering Scientist Associate
Applied Research Laboratories, University of Texas at Austin

2021 - 2022